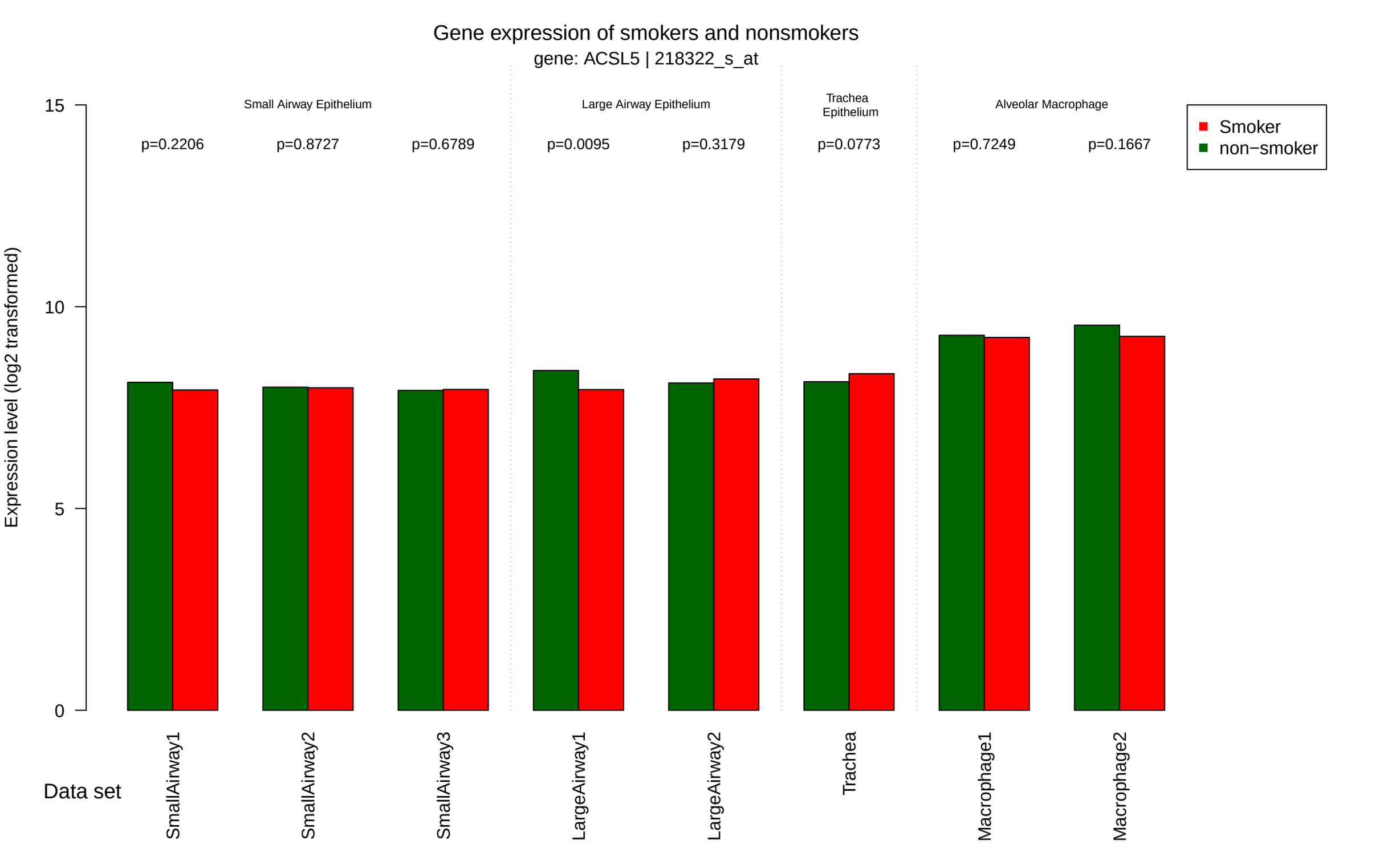
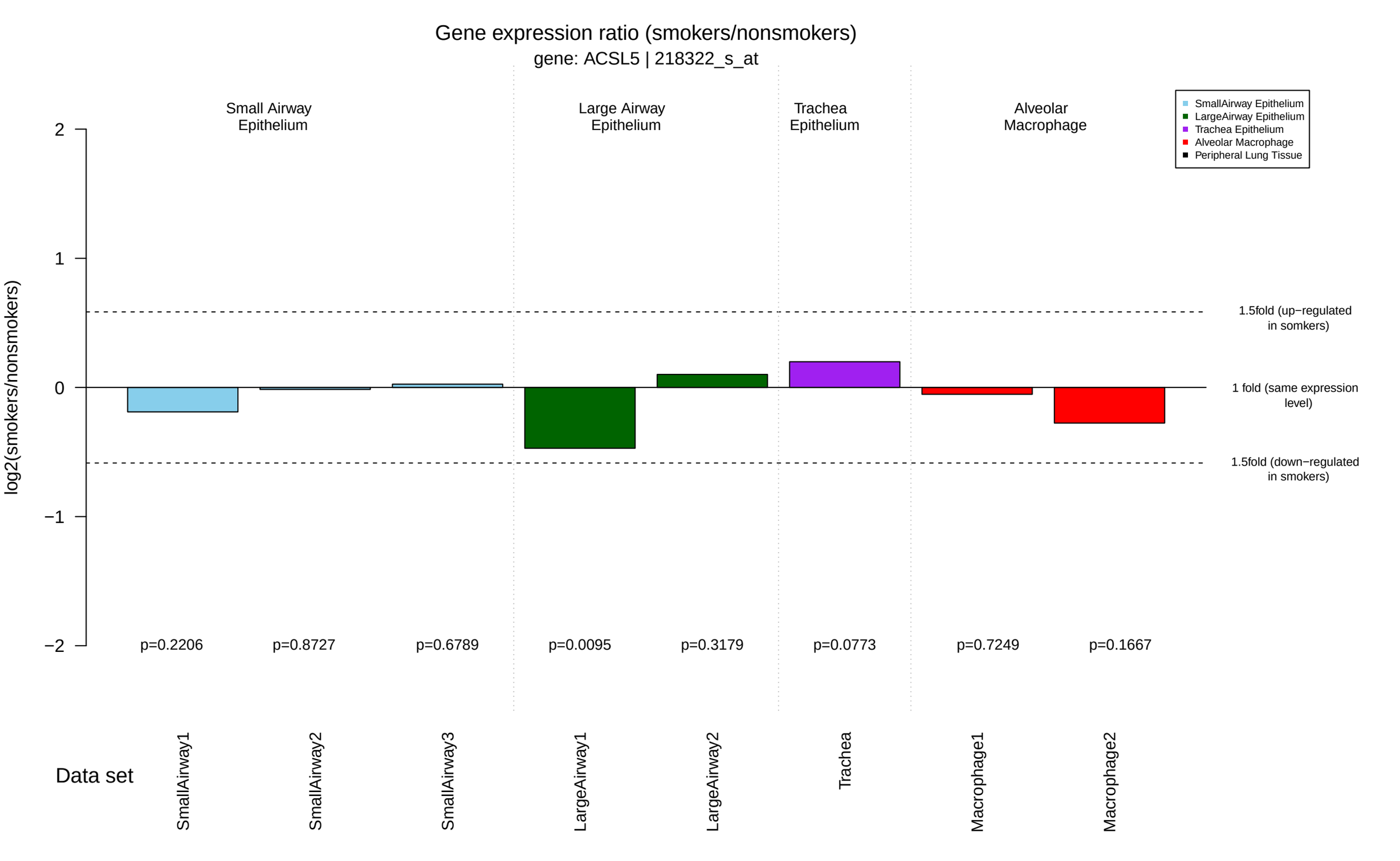
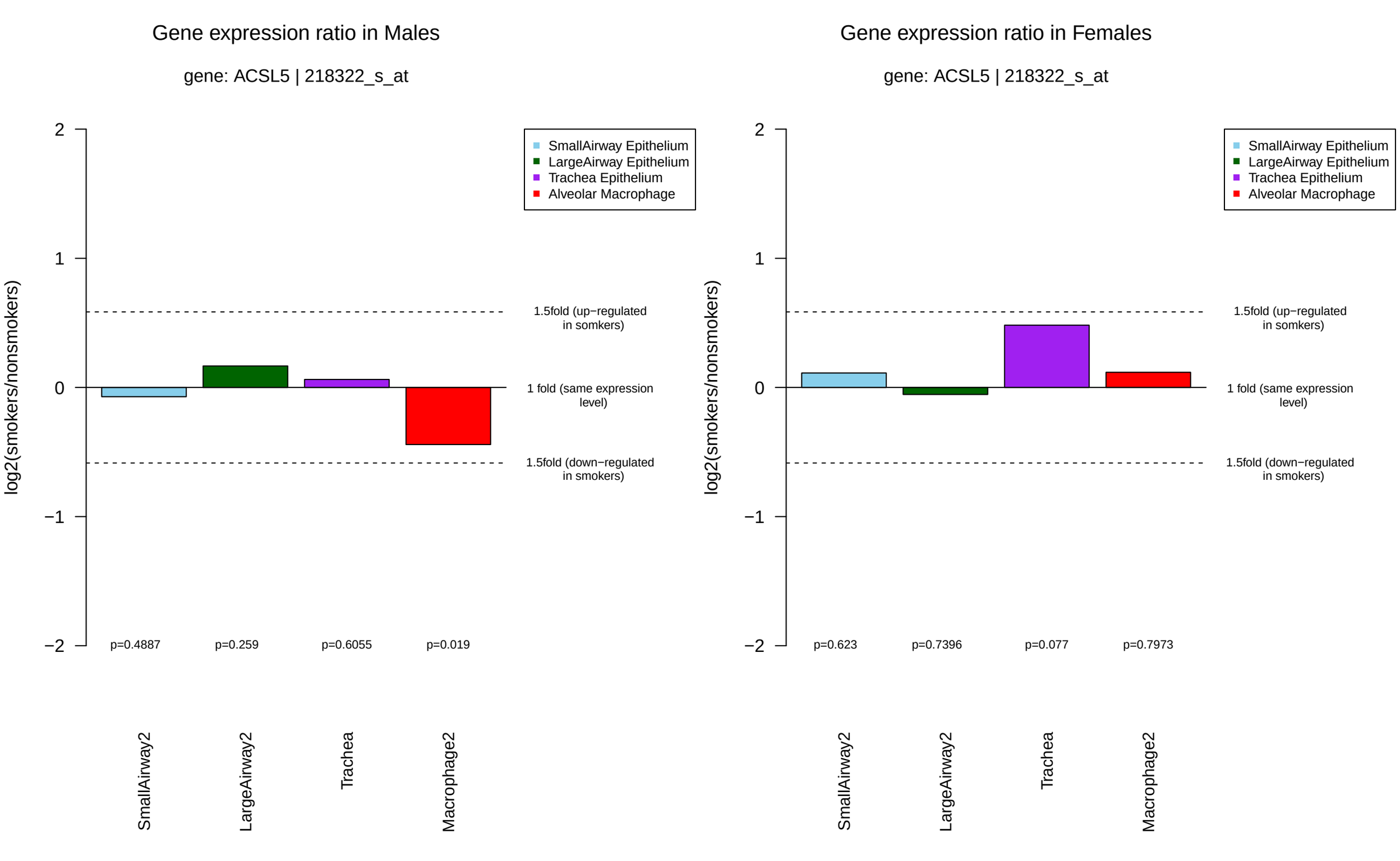
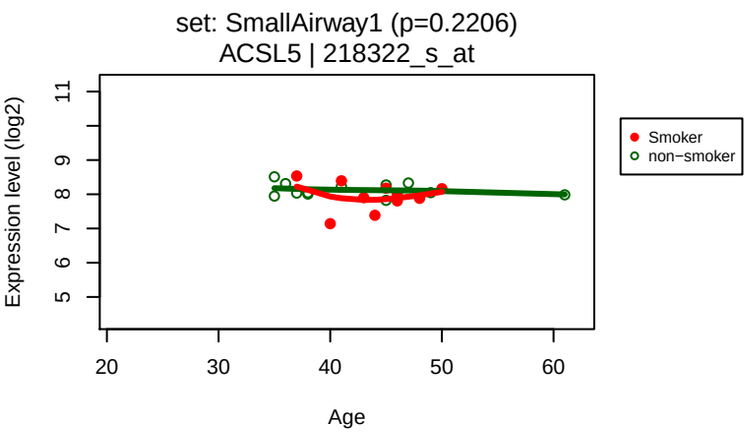
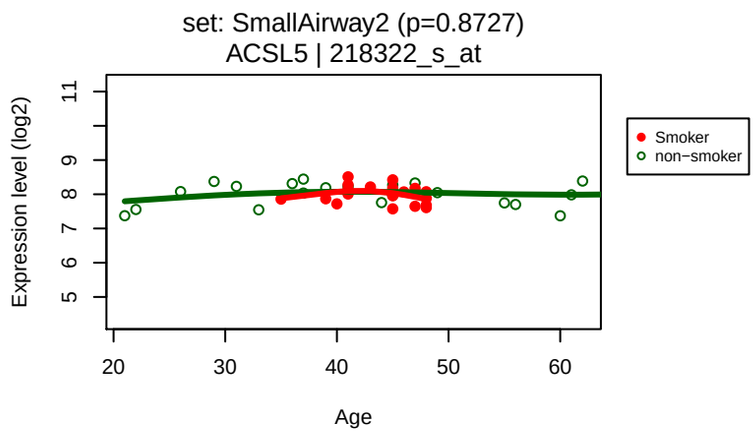
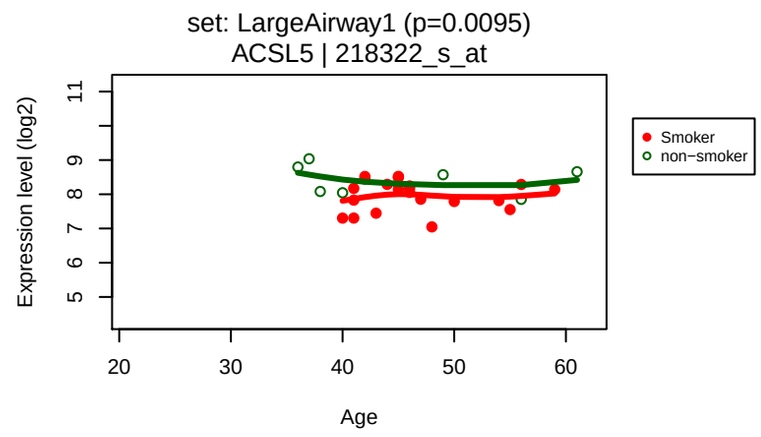

Gene expression of smokers and nonsmokers
gene: ACSL5 | 218322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2206
p=0.8727
p=0.6789
p=0.0095
p=0.3179
p=0.0773
p=0.7249
p=0.1667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

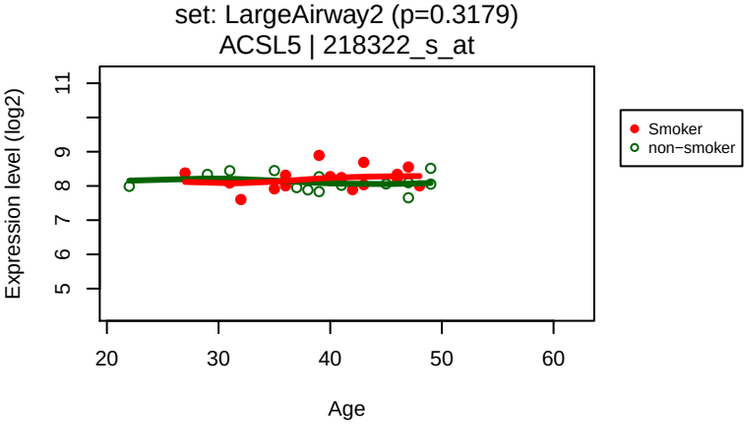

Gene expression ratio (smokers/nonsmokers)
gene: ACSL5 | 218322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2206
p=0.8727
p=0.6789
p=0.0095
p=0.3179
p=0.0773
p=0.7249
p=0.1667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

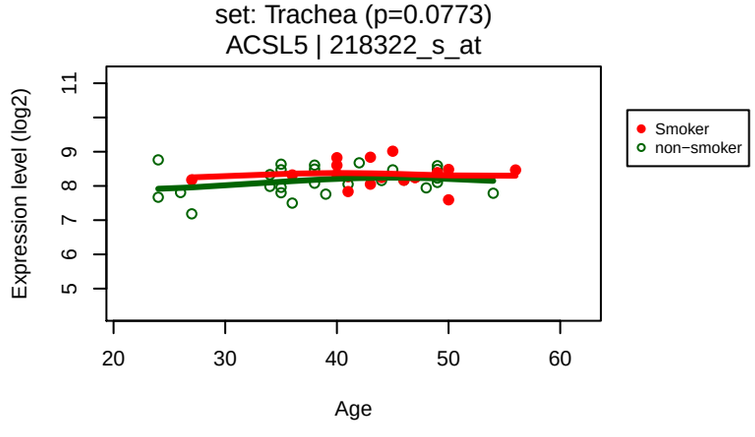

Gene expression ratio in Males
gene: ACSL5 | 218322_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4887
p=0.259
p=0.6055
p=0.019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSL5 | 218322_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.623
p=0.7396
p=0.077
p=0.7973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

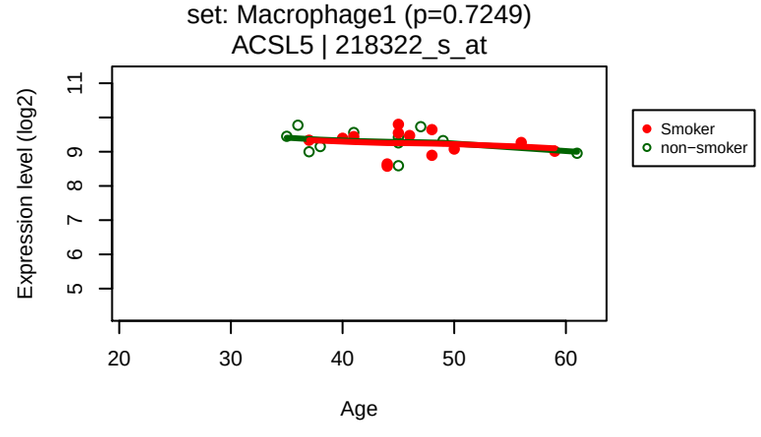

set: SmallAirway1 (p=0.2206)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker

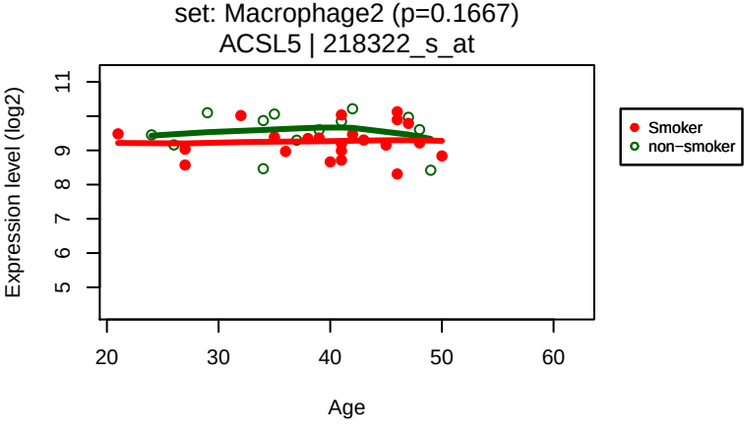

set: SmallAirway2 (p=0.8727)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker

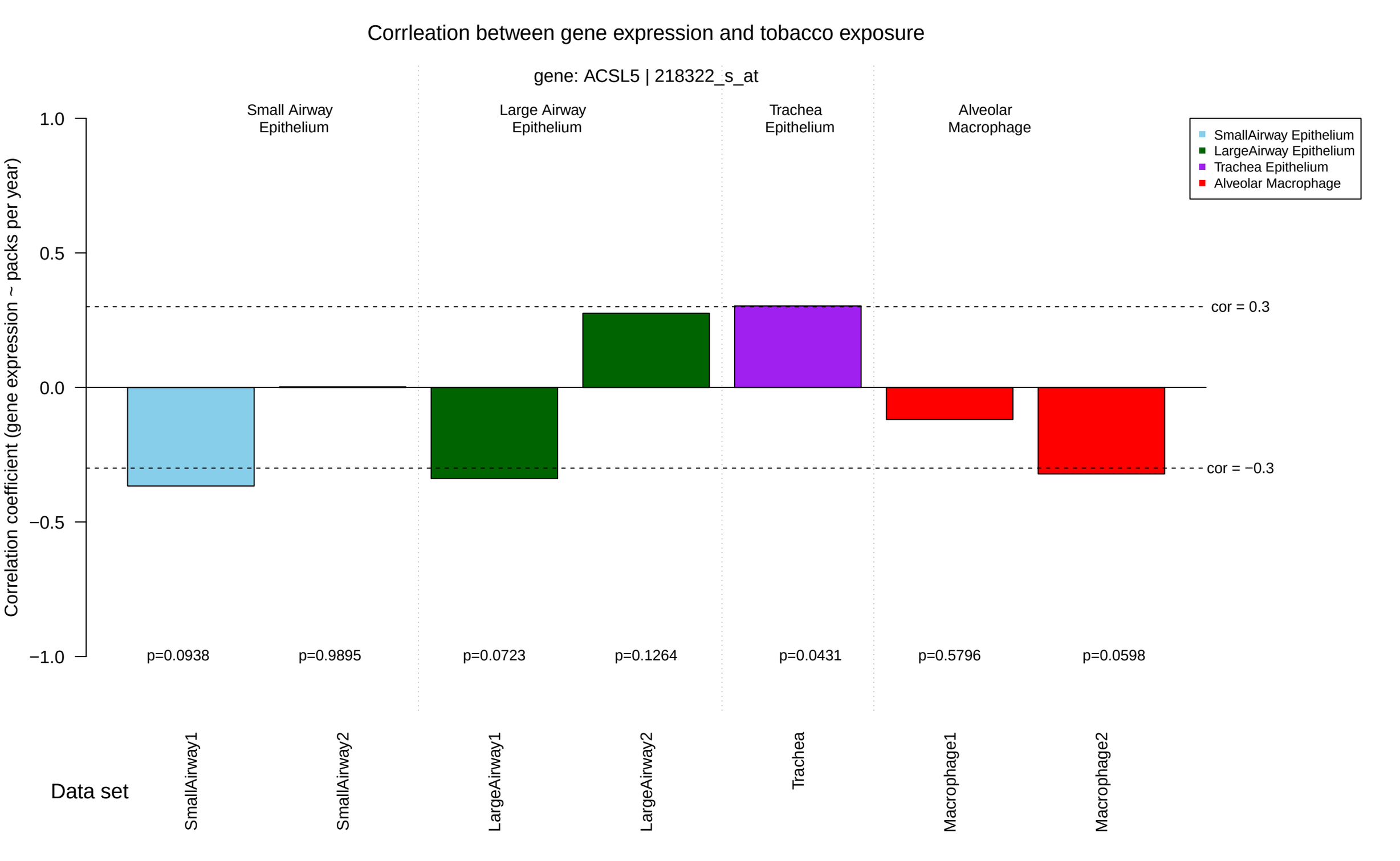

set: LargeAirway1 (p=0.0095)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3179)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0773)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7249)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1667)
ACSL5 | 218322_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSL5 | 218322_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0938
p=0.9895
p=0.0723
p=0.1264
p=0.0431
p=0.5796
p=0.0598
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2